



SYDE252 - W05A - Stability
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External Stability - Bounded-Input/Bounded-Output

For a bounded input (an input signal which **never** has a magnitude = infinity) do we get a bounded output (an output signal which **never** has a magnitude = infinity).

For our LTIC systems...

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau$$

Therefore:

$$|y(t)| \leq \int_{-\infty}^{\infty} |h(\tau)||x(t - \tau)|d\tau$$

Because $x(t)$ is bounded:

$$|x(t - \tau)| \leq K_1 < \infty$$

For BIBO Stability:

Then:

$$|y(t)| \leq K_1 \int_{-\infty}^{\infty} |h(\tau)|d\tau \implies \int_{-\infty}^{\infty} |h(\tau)|d\tau < \infty$$

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For our LTIC systems...

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

...if $h(t)$ is absolutely integrable, the system is BIBO stable.

Internal Stability - Asymptotic Stability

If, in the absence of an external input, a system remains in a particular state (or condition) indefinitely, then that state is an equilibrium state of the system.

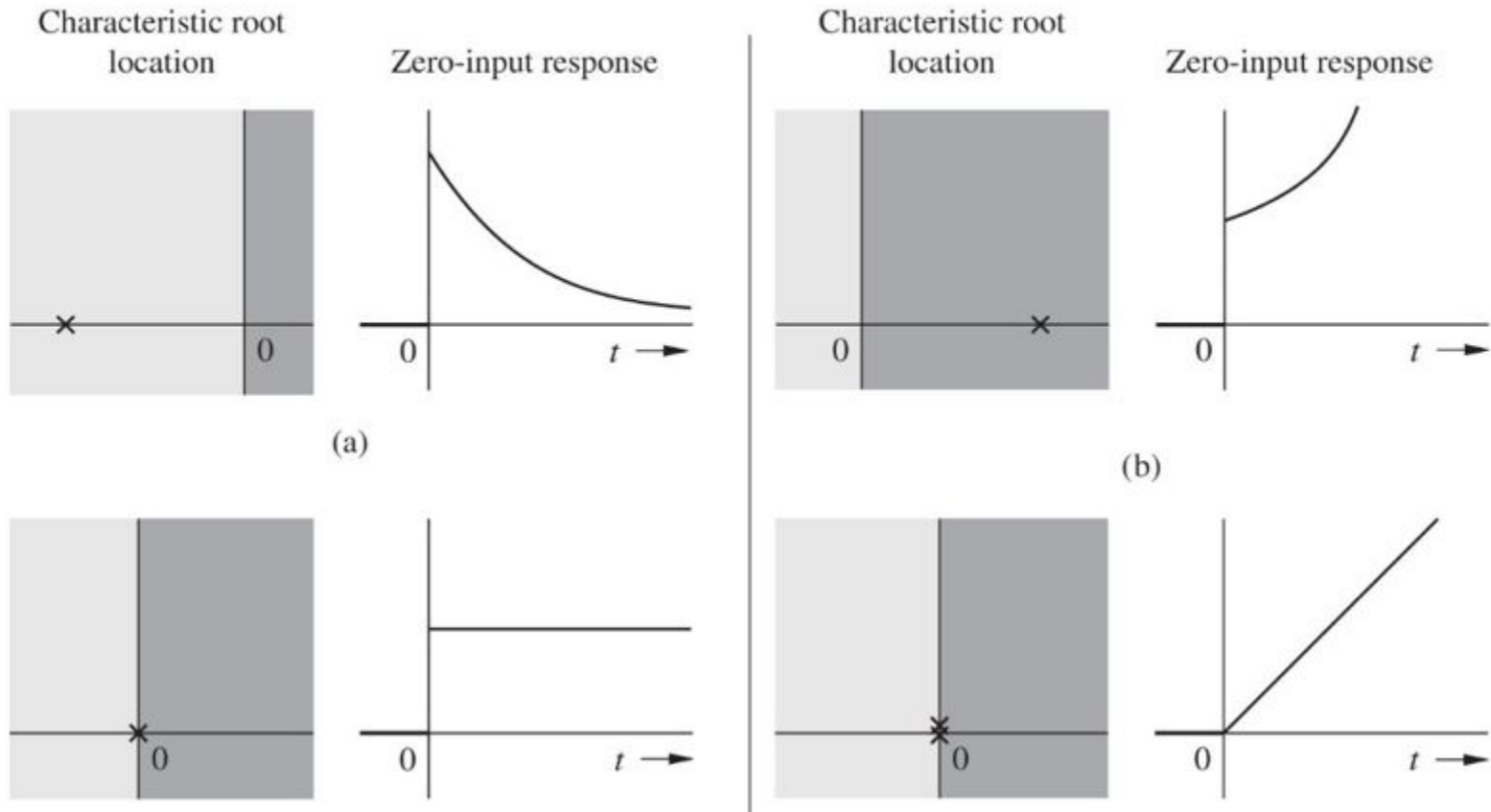
If we apply some IC's, but zero input, and the system returns to the equilibrium state, then it is ***internally stable (or asymptotically stable)***.

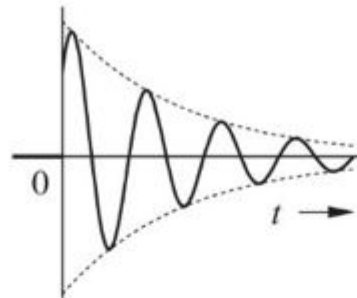
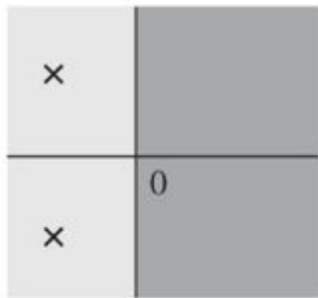
If it grows indefinitely it is ***unstable***.

If it doesn't grow indefinitely, but also doesn't return to the equilibrium state, it is ***marginally stable***.

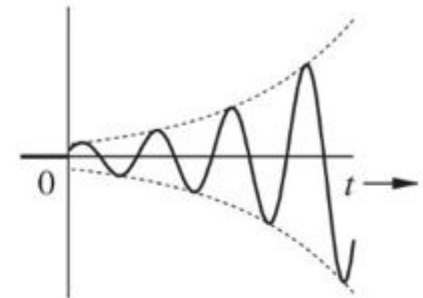
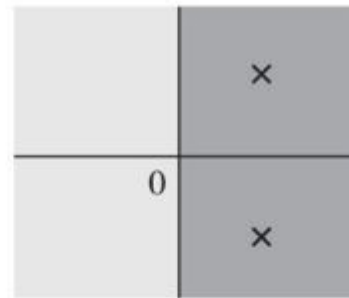
LTIC Systems:

1. Asymptotically stable iff all the characteristic roots are in the LHP (left hand plane, the real part < 0). The roots may all be unique or repeated.
2. Unstable iff one or both of the following are true:
 - a. At least one root is in the RHP (right hand plane, the real part > 0).
 - b. There are repeated roots on the imaginary axis (the real part $= 0$).
3. Marginally stable iff there are no roots in the RHP and there are some unrepeatd roots on the imaginary axis.

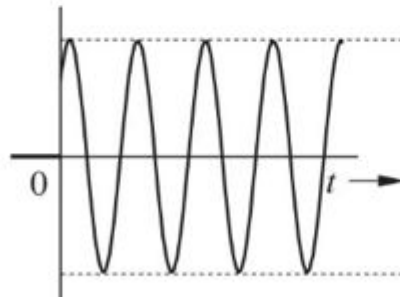
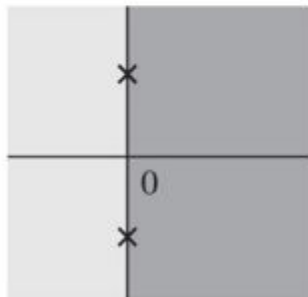




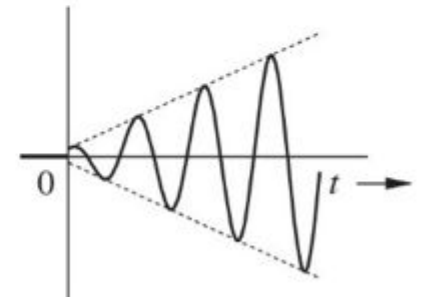
(e)



(f)



(g)



(h)

This is the s-plane - we'll talk more about this once we start on Laplace Transforms. Poles and zeros...

Laplace is coming.

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